

Truly Innovative **TECH**

Amongst numerous press releases of new products received by us, these are the ones we found worthy of the title *Truly Innovative Tech*

Wi-Fi SoCs covering long range, area, and volume

The Wi-Fi HaLow system-on-chip (SoC) claim to be the industry's smallest, fastest, and lowest-power IEEE 802.11ah-compliant SoCs. The MM6104 can sup-

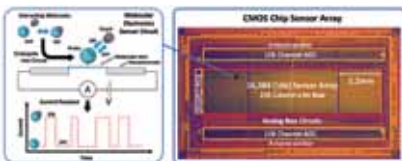


port 1, 2, and 4MHz channel bandwidth while the higher-performance MM6108 SoC supports 1, 2, 4, and 8MHz bandwidth and is capable of delivering tens of Mbps throughput to support streaming HD video. These Wi-Fi HaLow SoCs are thus capable of providing 10x the range, 100x the area, and 1000x the volume of traditional Wi-Fi solutions.

Morse Micro
www.morsemicro.com

Molecular electronics chip for biosensing

Realising a 50-year old vision of putting molecules into chips, the Roswell ME Platform is the first molecular electron-



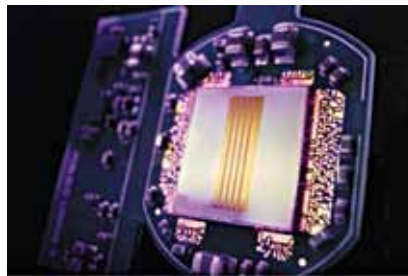
ics chip that has been developed for biosensing applications. It integrates single molecules into electronic circuits acting as sensor elements to create the first fully scaled biosensors on standard semiconductor chips. The sensor trans-

lates the dynamic process of molecular interactions into electrical measurements in real time, whose targets are programmed according to the particular molecule wired into the chip, providing for a programmable and universal biosensor platform.

Roswell Biotechnologies
www.roswellbiotech.com

Lidar SoC with on-chip signal processing

The L2X, which is the most powerful lidar system on chip (SoC) that has been created, features a highly sensitive SPAD photodetector combined with on-chip



digital signal processing that is capable of counting over one trillion photons per second and delivering double the data rate of the prior system on chip, while maintaining the same small size and low power draw. Its integration in the company's proprietary OS digital lidar sensors enables it to withstand high levels of shock, vibration, solar interference, water, and dust while reliably providing high-quality data.

Ouster
www.ouster.com

Echolocation sensor for 3D collision avoidance

The new echolocation sensor for 3D collision avoidance, called ECHO ONE DK, is based on ultrasonic technology, which mimics a bat's echolocation techniques. Unlike existing sensor technologies that can be negatively impacted by lighting

conditions or moisture (such as fog), it uses echolocation to generate robust, real-time 3D echo location data to guide

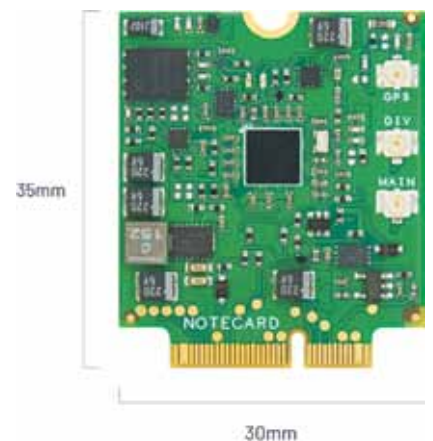


autonomous systems across a variety of applications. The performance of this 3D ultrasonic sensor is particularly robust in harsh environments (featuring the IP67 protection rating), as it is dirt tolerant and not affected by dust, smoke and mist.

TOPOSENS
www.toposens.com

SoM for secure, low-power cellular connectivity

The Notecard offers an easy way to add wide-range connectivity thanks to GNSS/GPS and global cellular over LTE-M, NB-IoT, or Cat-1, to any MCU



or SBC prototype to an existing production app. Since the Notecard comes with 500MB data and ten years of